Name_____

Photosynthesis: Making Energy

Chloroplasts

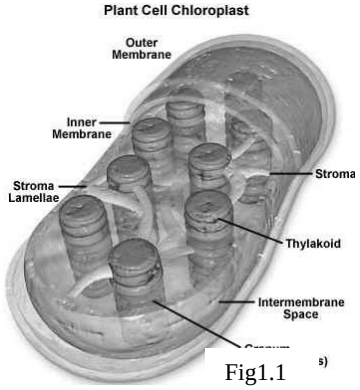


Fig1.1

Photosynthesis is a process in which sunlight energy is used to make glucose. The site of photosynthesis is in the **chloroplast** (Fig 1.1) – an organelle found in the leaves of green plants. The main functions of chloroplasts are to produce food (**glucose**) during **photosynthesis**, and to store food energy. Chloroplasts contain the pigment, *chlorophyll*. Chlorophyll absorbs most of the colors in the color spectrum, and reflects only green and yellow wavelengths of light. This is why we see leaves as green or yellow – because these colors are reflected into our eyes.

1. What is photosynthesis? _____
   _____
2. Where does photosynthesis occur? _____
3. What are chloroplasts and where are they found? _____
   _____
4. What are the two main functions of chloroplasts? _____
5. Why doe most leaves appear green? _____
6. What is the primary pigment found in the chloroplast? _____

Photosynthesis

Glucose is another name for sugar. The molecular formula for glucose is $C_6H_{12}O_6$. Plants make sugar by using the energy from sunlight to transform CO_2 from the air with water from the ground into glucose. This process, called photosynthesis occurs in the chloroplast of the plant cell. During this process, oxygen (O_2) is created as a waste product and is released into the air for us to breath. The formula for photosynthesis is:

(reactants) (products)

$$CO_2 + H_2O + \text{sunlight} \text{ ----> } C_6H_{12}O_6 + O_2$$

This formula says that carbon dioxide + water molecules are combined with the energy from sunlight to produce sugar and oxygen. The reactants in photosynthesis (what is used) are CO_2, water and sun. The plant gets water from the ground through its roots. The plant collects carbon dioxide from the air. Much of the carbon dioxide comes from living organisms that exhale (breath it out) it, but some also comes from factory smokestacks and car fumes.

7. What is the formula for photosynthesis? _____
8. What three things are used to make glucose in photosynthesis? _____
   _____
9. Where does the water come from? _____
10. Where does the water enter the plant? _____
11. Name 3 some sources of CO_2. _____
12. What type of energy does the plant use to convert CO_2 and H_2O into sugar?
    _____

The products are **glucose** and **oxygen**. The glucose produced is used by the plant for energy and growth. We also use this glucose by eating plants. The oxygen produced is released into the air for us to breath. Photosynthesis is essential for all life on earth, because it provides food and

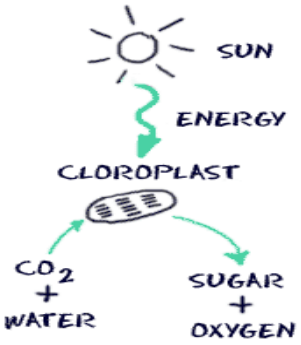
Name_____

oxygen. Plants are considered autotrophs because unlike us humans, they can make their own food using this process.

13. What is produced in photosynthesis? _____
14. What is the glucose used for? _____
15. What is the oxygen used for? _____

16. Here are three different ways to visualize the photosynthesis reaction: Is it easier for you to understand the reaction by using pictures, words, or symbols (see above)? Why?

_____
_____

Photosynthesis in pictures	Photosynthesis in words	Photosynthesis in symbols
	Carbon dioxide and water combine with sunlight to create oxygen and glucose.	$CO + H_2O \xrightarrow{\text{light}} C_6H_{12}O_6 + O_2$

<u>Essential Question</u>: Describe, using scientific terms, how plants turn sunlight into energy? Make sure to refer to the chemical equation to photosynthesis and discus the reactants and products.

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_____
_____
_____